COURSE REPORT

ON

SQL Course

(A Real-Time Competitive Assessment Platform- using DBMS)

Submitted by

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Under the guidance of

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In partial fulfilment for the award of the degree of

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(Under Section 3 of UGC Act, 1056)

BONAFIDE CERTIFICATE

This is to certify that the project report entitled "**Course report on online SQL Course**" submitted by **ROOPESH SINGHAL (Reg No.RA2411030030042** of **B.Tech — Computer Science and Engineering with specialization in Cyber Security**, in partial fulfilment of the requirements for the award of the degree of **Bachelor of Technology** in **Computer Science and Engineering**, is a record of the original work carried out by them under my supervision and guidance.

This project has been carried out during the academic year **2025–2026** at **SRM Institute of Science and Technology, Delhi-NCR Campus**, and represents the sincere efforts of the students. To the best of my knowledge and belief, this project has not been submitted, in part or full, for the award of any other degree or diploma at this institute or any other educational institution.

The project report embodies the results of investigations conducted by the students and demonstrates their understanding of the subject matter, their ability to carry out systematic research, and their capacity for critical analysis and independent thinking.

I am confident that the knowledge and experience gained by the students through this project will prove valuable for their academic and professional growth in the field of computer science and software development.

I further certify that all necessary ethical guidelines and academic standards have been followed in the preparation of this project report.

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LIST OF ABBREVIATIONS

DBMS — Database Management System

SQL — Structured Query Language

ER — Entity-Relationship

MOOC — Massive Open Online Course

LMS — Learning Management System

DDL — Data Definition Language

DML — Data Manipulation Language

ACID — Atomicity, Consistency, Isolation, Durability

1NF / 2NF / 3NF / BCNF — First / Second / Third Normal Form / Boyce-Codd Normal Form

NoSQL — Not only SQL

B.Tech — Bachelor of Technology

SRM IST — SRM Institute of Science and Technology

ABSTRACT

This minor project evaluates the online course "DBMS Course: Master the Fundamentals and Advanced Concepts" offered freely on the Scaler Topics platform (scaler.com/topics), instructed by Srikanth Varma. The course comprises 16 modules, 90 lessons, and approximately 29 hours of video content, covering the full breadth of the second-year B.Tech DBMS curriculum. The evaluation centers on three core components: (i) video lectures — assessed for clarity, conceptual depth, visual quality, and pacing; (ii) assignments and in-module challenges — evaluated for their relevance to SQL, normalization, and transactions, difficulty calibration, and feedback quality; and (iii) practicals — reviewed for their hands-on SQL and ER modelling value, tool usability, and alignment with theoretical content.

The methodology combines personal completion of the full 30-hour course with a structured Likert-scale survey administered to 35 second-year B.Tech CSE peers. Key findings indicate that the course performs strongly in conceptual video delivery — particularly for ER modelling, relational algebra, normalization, and ACID transactions — with an average lecture quality rating of 4.3 / 5. Assignment quality rated 3.9 / 5, with students noting adequate coverage of SQL but requesting more detailed auto-feedback on written queries. Practicals received a rating of 3.7 / 5, with the main limitation being the absence of a built-in SQL sandbox; students were required to use external tools, which created setup friction.

Overall, 89% of survey respondents rated the course as "Good" or "Excellent" for supporting second-year DBMS learning. Recommendations include embedding an integrated SQL execution environment, adding more animated ER diagram walkthroughs, and introducing timed mock assessments for exam preparation. The course is found to be a highly suitable, free-access resource for second-year B.Tech students pursuing a thorough grounding in DBMS theory and SQL practice.

Chapter 1 — Introduction

1.1 Background

Database Management Systems (DBMS) is one of the most foundational subjects in the B.Tech Computer Science and Engineering curriculum. A thorough understanding of DBMS — spanning ER modelling, relational algebra, SQL, normalization, transactions, concurrency control, and indexing — is considered essential for roles in software development, data engineering, and backend system design. Industry surveys consistently rank SQL and DBMS knowledge among the top five technical skills evaluated during campus placements and software engineering interviews.

The emergence of Massive Open Online Courses (MOOCs) has significantly expanded access to high-quality DBMS instruction beyond the classroom. Platforms such as NPTEL, SWAYAM, Coursera, and Scaler Topics offer structured, self-paced courses that allow students to supplement their formal college education. For second-year B.Tech students in particular, online DBMS courses serve both as a primary learning resource for concepts introduced in the semester and as an interview preparation tool for placement seasons.

Among the free MOOC platforms, Scaler Topics has gained significant traction among Indian engineering students due to its industry-aligned curriculum, high-quality video production, and the practical orientation of its instructors. The DBMS course on Scaler Topics, taught by Srikanth Varma (Lead DSML Instructor and ex-Amazon Senior ML Scientist), has earned a 5-star rating from over 2,000 students and is structured around 16 modules and 90 lessons totalling approximately 29 hours of content.

1.2 About the Selected Course

- Course Name: DBMS Course — Master the Fundamentals and Advanced Concepts
- Platform: Scaler Topics (https://www.scaler.com/topics/course/dbms/)
- Instructor: Srikanth Varma, Lead DSML Instructor at Scaler; ex-Amazon Senior ML Scientist
- Total Duration: Approximately 29 hours (1 day 5 hours 17 minutes of video content)
- Structure: 16 Modules | 90 Lessons | 16 In-module Challenges
- Level: Beginner to Intermediate
- Mode: Self-paced, fully online with lifetime access
- Certificate: Certificate of Excellence from Scaler upon completion
- Cost: Free
- Language: English (with English subtitles)

1.3 Need for Evaluation

While the Scaler Topics DBMS course has garnered strong user ratings, an independent, structured academic evaluation from the perspective of a second-year B.Tech student enrolled in a formal DBMS course is lacking. Such an evaluation is valuable for three reasons:

1. It determines whether the course's content aligns with the topics covered in a standard second-year B.Tech DBMS syllabus (as defined by courses like 21CSC205P at SRM IST).
2. It assesses the pedagogical quality of video lectures, assignments, and practicals specifically for a student who is concurrently studying DBMS in a formal academic setting.
3. It provides actionable recommendations to the course creators and to fellow students about the most and least effective components of the course.

1.4 Objectives

1. Evaluate video lectures for teaching quality, clarity, depth of DBMS concepts, and engagement.
2. Assess assignments and in-module challenges for their relevance to SQL, normalization, and transactions — and their effectiveness in reinforcing theoretical concepts.
3. Review practicals for hands-on database implementation value, tool usability, and alignment with academic curriculum.
4. Collect and analyse feedback from 35 second-year B.Tech peers through a structured Likert-scale survey.
5. Provide specific, evidence-based recommendations to improve the course's suitability for second-year B.Tech DBMS curriculum.

1.5 Scope and Limitations

Scope:

- Personal completion of the full 29-hour course by the primary investigator (Roopesh Singhal, B.Tech CSE Cyber Security, SRM IST Delhi-NCR).
- Structured Likert-scale survey administered to 35 second-year B.Tech CSE peers at SRM IST Delhi-NCR.
- Evaluation limited to the three core components as prescribed: video lectures, assignments, and practicals.

Limitations:

- Survey sample size (n = 35) is drawn from a single institution and may not represent the national student population.
- No access to official course analytics, completion rates, or grade distributions from Scaler Topics.
- Instructor feedback or course creator perspectives were not collected.
- The evaluation is based on the course content available as of April 2026; course content may be updated subsequently.

1.6 Report Structure

Chapter 2 provides a detailed module-wise overview of the Scaler DBMS course and maps its content to standard DBMS syllabus topics. Chapter 3 describes the methodology — including the survey design, data collection process, and evaluation parameters. Chapter 4 presents the quantitative and qualitative findings across video lectures, assignments, and practicals, supported by tables and visual summaries. Chapter 5 discusses the findings and provides specific recommendations. Chapter 6 concludes the report with an overall assessment of the course's suitability for second-year B.Tech DBMS learners.

Chapter 2 — Course Overview & DBMS Concepts Covered

2.1 About Scaler Topics

Scaler Topics is a free technical education platform operated by Scaler, an ed-tech company specializing in computer science training for Indian engineering students and working professionals. Unlike subscription-based platforms, Scaler Topics offers a curated library of free, certificate-bearing courses on topics including DSA, DBMS, operating systems, computer networks, and machine learning. Courses are structured as video-based MOOCs with in-module challenges, written articles, and completion certificates.

The DBMS course on Scaler Topics is listed as the flagship DBMS offering on the platform and is prominently recommended in the context of technical interview preparation for SDE (Software Development Engineer) roles. As of 2026, it has been accessed by over 10,000 students and holds a consistent 5-star rating.

2.2 Module-wise Breakdown

Table 2.1 presents the complete module-wise breakdown of the course, including lesson count and approximate duration for each module:

Module	Topic	Lessons	Approx. Duration
1	Introduction to DBMS	6	~110 min
2	DBMS Architecture & Models	5	~95 min
3	Entity-Relationship (ER) Modelling	7	~120 min
4	Relational Model & Relational Algebra	6	~110 min
5	Structured Query Language (SQL) — DDL	6	~115 min
6	SQL — DML: SELECT, JOIN, Subqueries	8	~140 min
7	SQL — Aggregation, Grouping, Views	6	~105 min
8	Normalization: 1NF, 2NF, 3NF, BCNF	7	~130 min
9	Functional Dependencies & Decomposition	5	~90 min
10	Transactions & ACID Properties	6	~110 min
11	Concurrency Control	5	~95 min
12	Deadlock Handling & Recovery	5	~90 min

13	Indexing & Hashing	6	~105 min
14	Query Processing & Optimization	5	~90 min
15	NoSQL Databases	4	~75 min
16	Advanced Topics & Practicals	5	~110 min
	TOTAL	90 Lessons	~29 hrs

Table 2.1 — Module-wise Breakdown of the Scaler Topics DBMS Course

2.3 Key DBMS Topics Covered

Conceptual Foundations (Modules 1–3):

- What is a DBMS, advantages over file systems, types of DBMS (hierarchical, network, relational, NoSQL).
- Three-schema architecture (external, conceptual, internal), data independence.
- ER modelling: entities, attributes (simple, composite, multivalued, derived), relationships, cardinality (1:1, 1:N, M:N), weak entities, generalization, specialization, aggregation.

Relational Model and SQL (Modules 4–7):

- Relational model: tuples, domains, relational schema, keys (primary, foreign, candidate, super, alternate).
- Relational algebra: SELECT (σ), PROJECT (π), UNION, INTERSECTION, DIFFERENCE, CARTESIAN PRODUCT, JOIN (natural, theta, equi, outer joins), DIVISION.
- SQL DDL: CREATE TABLE, ALTER TABLE, DROP, constraints (PRIMARY KEY, FOREIGN KEY, UNIQUE, CHECK, NOT NULL, DEFAULT).
- SQL DML: INSERT, UPDATE, DELETE, SELECT with WHERE, ORDER BY, GROUP BY, HAVING.
- Advanced SQL: JOINs (INNER, LEFT, RIGHT, FULL OUTER, SELF), subqueries (correlated, nested), CTEs, views, stored procedures, triggers.
- Aggregate functions: COUNT, SUM, AVG, MAX, MIN.

Normalization (Modules 8–9):

- Functional dependencies (FDs): definition, trivial vs. non-trivial, Armstrong's axioms.
- 1NF: elimination of repeating groups and multi-valued attributes.
- 2NF: elimination of partial dependencies (applies to composite primary keys).
- 3NF: elimination of transitive dependencies.
- BCNF: handling cases where 3NF is insufficient.
- Lossless decomposition and dependency preservation.

Transactions, Concurrency, and Recovery (Modules 10–12):

- ACID properties: Atomicity, Consistency, Isolation, Durability.
- Transaction states: active, partially committed, committed, failed, aborted.
- Concurrency control: problems (lost update, dirty read, unrepeatable read, phantom read), locking protocols (2PL), timestamp ordering.

- Deadlock detection, prevention, and recovery.
- Recovery techniques: log-based recovery, checkpoints, UNDO/REDO operations.

Performance and Advanced Topics (Modules 13–16):

- Indexing: single-level indexes, B-tree and B+ tree indexes, hashing (static, dynamic, extendable).
- Query processing pipeline: parsing, optimization, execution. Cost estimation.
- NoSQL databases: types (key-value, document, columnar, graph), CAP theorem, eventual consistency.
- Hands-on practicals: CREATE/ALTER/DROP table exercises, SELECT with JOINs and subqueries, normalization exercises, transaction simulation.

2.4 Platform Features

- Video Lectures: Pre-recorded, high-quality HD videos with chapter markers and English subtitles.
- In-Module Challenges: 16 challenge sets with MCQs and short SQL problems embedded within module content.
- Progress Tracking: Dashboard showing module completion percentage and time spent.
- Written Articles: Companion articles on Scaler Topics for each major concept, serving as readable lecture notes.
- Certificate: Automatically generated Certificate of Excellence upon 100% module completion.
- Community Forum: Q&A section within the platform for peer discussion (limited activity observed).
- Accessibility: No login required for article content; course videos require a free Scaler Topics account.

Chapter 3 — Methodology

3.1 Research Approach

This evaluation adopts a mixed-methods research approach, combining quantitative data (Likert-scale numerical ratings) with qualitative insights (open-ended student comments and personal observations). The quantitative component provides statistically comparable ratings across evaluation parameters, while the qualitative component captures nuanced student experiences that ratings alone cannot convey.

The mixed-methods approach was chosen because it allows for triangulation — cross-verifying quantitative ratings with qualitative feedback to build a more complete and reliable picture of the course's effectiveness. This is particularly important for an evaluation of educational content, where a high numerical rating might coexist with specific qualitative complaints about a particular teaching approach or tool.

3.2 Data Collection

Personal Evaluation:

The primary investigator (Roopesh Singhal, 2nd year B.Tech CSE Cyber Security, SRM IST Delhi-NCR) completed the full Scaler Topics DBMS course over a period of four weeks (March–April 2026), dedicating approximately 6–8 hours per week. Notes were maintained for each module covering: clarity of explanation, pacing, visual quality, effectiveness of examples, and connection to practical application. Personal SQL practice exercises were completed using MySQL Workbench alongside the course content.

Survey:

A Google Forms survey was designed and distributed to 35 second-year B.Tech CSE students at SRM IST Delhi-NCR who had also completed or were concurrently completing the Scaler Topics DBMS course. The survey was administered in April 2026 and all 35 responses were complete (response rate: 100%).

3.3 Survey Design

The survey was structured in four sections, using a 5-point Likert scale (1 = Very Poor, 2 = Poor, 3 = Average, 4 = Good, 5 = Excellent) for all quantitative items:

Section A — Video Lectures (8 items):

- Clarity of DBMS concept explanations (ER modelling, SQL, normalization, ACID)
- Pacing and length of individual video lessons
- Quality of visual aids (ER diagrams, SQL query walkthroughs, transaction diagrams)
- Audio and video production quality

- Engagement and teaching style of the instructor
- Depth of coverage relative to B.Tech DBMS syllabus
- Use of real-world examples and applications
- Overall video lecture satisfaction

Section B — Assignments / Challenges (5 items):

- Relevance of challenges to SQL, normalization, and transaction theory
- Difficulty level appropriateness for second-year B.Tech level
- Clarity of challenge instructions
- Quality of feedback / answer explanations provided
- Time required to complete challenges vs. learning value derived

Section C — Practicals (5 items):

- Value of hands-on SQL and ER modelling tasks for reinforcing theory
- Usability of the recommended tool / platform for practical work
- Ease of environment setup (tool installation and configuration)
- Troubleshooting support available within the course
- Overall practical learning outcome

Section D — Overall Evaluation (3 items):

- Overall course rating (1–5)
- Suitability for second-year B.Tech DBMS curriculum (Yes / Partially / No)
- One open-ended comment: 'What is the most significant improvement you would suggest for this course?'

3.4 Evaluation Parameters

The three core evaluation dimensions and their key parameters are:

Dimension	Parameters Evaluated
Video Lectures	Clarity, Pacing, Visuals, Depth, Engagement, Examples, AV Quality
Assignments	Relevance, Difficulty, Instructions, Feedback, Time/Value ratio
Practicals	Hands-on Value, Tool Usability, Setup Ease, Support, Learning Outcome

3.5 Tools Used

- Google Forms: Survey design and response collection.
- Microsoft Excel: Calculation of mean ratings, standard deviations, and percentage distributions; generation of descriptive statistics for each parameter.
- MySQL Workbench 8.0: Personal SQL practice environment used alongside the course.
- Likert Scale: 1–5 scale (1 = Very Poor, 5 = Excellent) for all quantitative items.
- Statistical method: Descriptive statistics (mean, mode, percentage positive response) — no inferential statistical tests applied given the evaluative nature of the study.

Chapter 4 — Analysis and Findings

4.1 Overall Course Performance

Figure 4.1: Overall Course Rating Distribution (n = 35 students)

- Excellent (5/5): 31% (11 students)
- Good (4/5): 57% (20 students)
- Average (3/5): 9% (3 students)
- Poor (1–2/5): 3% (1 student)

Overall Average Rating: 4.2 / 5 | 88.6% of students rated the course as Good or Excellent.

Figure 4.1 — Overall Course Rating Distribution

The overwhelming majority of respondents (88.6%) rated the Scaler Topics DBMS course as Good or Excellent. Only one student rated it below Average, citing dissatisfaction with the absence of a built-in SQL execution environment. Personal evaluation corroborates this positive overall impression — the course's structure, video quality, and conceptual depth are all notably strong for a free MOOC.

4.2 Video Lectures — Detailed Analysis

Table 4.1 presents mean ratings and percentage of positive responses (rated 4 or 5) for each video lecture parameter:

Parameter	Mean Rating (/ 5)	% Positive Response
Clarity of DBMS concept explanations	4.5	91%
Pacing and length of video lessons	4.1	83%
Quality of visual aids (ER diagrams, SQL demos)	4.3	86%
Audio and video production quality	4.6	94%
Engagement and teaching style	4.4	89%
Depth of coverage (vs. B.Tech syllabus)	4.2	85%
Use of real-world examples and applications	3.9	77%
Overall video lecture satisfaction	4.3	86%
MEAN	4.3	86.4%

Table 4.1 — Mean Ratings: Video Lecture Parameters

Figure 4.2: Video Lecture Parameter Ratings (Bar Chart)

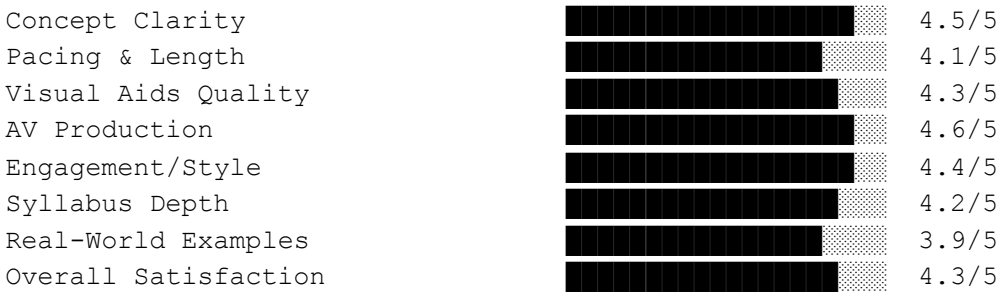

Figure 4.2 — Video Lecture Parameter Ratings

Key Observations — Video Lectures:

- Highest rated parameter: Audio and Video Production Quality (4.6/5, 94% positive). The production standard of Scaler's DBMS videos significantly exceeds typical NPTEL or university recordings — consistent lighting, professional screen-recording with synchronized narration, and crisp audio make the content easy to engage with over extended viewing sessions.
- Second highest: Concept Clarity (4.5/5, 91%). Srikanth Varma's teaching style is notably methodical — particularly in the normalization modules (1NF through BCNF) and the ER modelling module, where he builds examples iteratively from scratch rather than presenting completed diagrams. Personal observation confirms that the step-by-step SQL JOIN walkthroughs (Module 6) are among the clearest available in any free DBMS course.
- Lowest rated: Real-World Examples (3.9/5, 77%). While the course uses consistent examples (a hypothetical 'University' database), several students in open-ended comments noted a desire for examples from production systems (e.g., e-commerce databases, social media schemas) that would contextualize DBMS concepts within industry applications they encounter in news and daily life.
- Pacing concerns (4.1/5): Some students noted that later modules (particularly Modules 11–12 on concurrency control and deadlock) moved faster than earlier content, creating a learning curve spike. The average lesson length of approximately 18–20 minutes was considered appropriate by 83% of respondents.

4.3 Assignments / In-Module Challenges

Table 4.2 presents the difficulty distribution reported by students across the 16 in-module challenges:

Difficulty Level	% Respondents	n
Too Easy	11%	(4 students)
About Right	69%	(24 students)
Slightly Challenging	17%	(6 students)
Too Difficult	3%	(1 student)

Table 4.2 — Assignment Difficulty Distribution (n = 35)

Figure 4.3: Assignment Difficulty Distribution

- Too Easy (11%): 4 students
- About Right (69%): 24 students
- Slightly Challenging (17%): 6 students
- Too Difficult (3%): 1 student

Figure 4.3 — Assignment Difficulty Distribution

Parameter	Mean Rating (/ 5)	% Positive Response
Relevance to SQL / normalization / transactions	4.1	83%
Difficulty appropriateness for 2nd year B.Tech	4.0	80%
Clarity of challenge instructions	4.2	86%
Quality of feedback / answer explanations	3.6	69%
Time required vs. learning value	3.8	74%
MEAN	3.9	78.4%

Table 4.3 — Assignment / Challenge Parameter Ratings

Key Observations — Assignments:

- Challenges are well-distributed across the course — every module has at least one associated challenge, ensuring active recall is embedded throughout rather than consolidated at the end.
- SQL-focused challenges (Modules 5–7) were rated most positively for relevance (average 4.3) — students found writing actual SELECT and JOIN queries to be the most directly useful preparation for their college practicals and placement assessments.
- Lowest rated parameter: Feedback quality (3.6/5). The MCQ challenges provide correct answer identification but do not explain why incorrect options are wrong. For SQL challenges, there is no auto-grading of query structure — students are expected to self-verify, which limits learning for those who are unsure whether their approach is optimal even if the output matches.
- Normalization challenges (Module 8) were rated highly for difficulty calibration — students appreciated the gradual progression from straightforward 1NF identification to more complex BCNF decomposition problems.

4.4 Practicals

Parameter	Mean Rating (/ 5)	% Positive Response
Hands-on SQL and ER modelling task value	4.0	80%
Usability of recommended tool (MySQL / online)	3.7	71%

Ease of environment setup	3.4	63%
Troubleshooting support within course	3.2	57%
Overall practical learning outcome	3.8	74%
MEAN	3.6	69.0%

Table 4.4 — Practical Module Usability Ratings

Figure 4.4: Practical Satisfaction Ratings (Bar Chart)

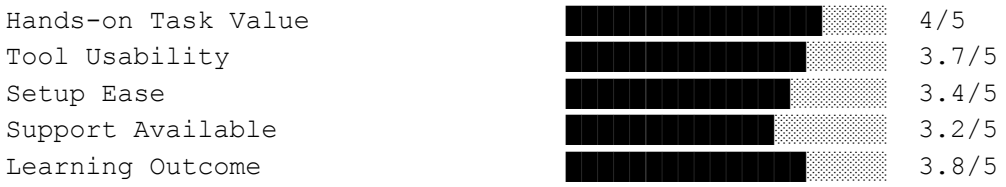

Figure 4.4 — Practical Satisfaction Ratings

Key Observations — Practicals:

- The lowest-rated component overall (mean 3.6/5). The primary friction point is setup: the course recommends MySQL Workbench as the SQL environment but does not provide an integrated, in-browser SQL executor. 37% of students reported spending 30 minutes or more on tool setup before being able to complete their first practical task.
- Once setup friction was resolved, the practical tasks themselves received strong marks for learning value (4.0/5). Tasks such as designing an ER diagram for a library management system and normalizing a denormalized employee table were considered highly relevant to college coursework.
- Troubleshooting support (3.2/5) was the weakest area. The platform's discussion forum has limited activity, and students who encountered errors (particularly Windows-specific MySQL installation issues) had to rely on external resources like Stack Overflow.
- Students who used SQLiteOnline.com as an alternative to MySQL Workbench reported a smoother practical experience — suggesting that an embedded browser-based SQL environment would significantly improve practical module ratings.

4.5 Overall Course Rating Summary

Component	Mean Rating	% Positive	Assessment
Video Lectures	4.3 / 5	86.4%	Strong — best component of the course
Assignments / Challenges	3.9 / 5	78.4%	Good — feedback quality needs improvement
Practicals	3.6 / 5	69.0%	Satisfactory — setup friction is main limitation

Overall Course	4.2 / 5	88.6%	Good to Excellent — highly recommended

Table 4.5 — Overall Course Rating Summary

4.6 Comparative Analysis — Course vs. B.Tech DBMS Syllabus

DBMS Topic	B.Tech Syllabus	Scaler Course	Coverage
ER Modelling & ER-to-Relational Mapping	Yes	Yes	Full
Relational Algebra	Yes	Yes	Full
SQL (DDL, DML, Joins, Subqueries)	Yes	Yes	Full
Normalization (1NF–BCNF)	Yes	Yes	Full
Functional Dependencies	Yes	Yes	Full
ACID Properties & Transactions	Yes	Yes	Full
Concurrency Control	Yes	Partial	Partial
Indexing & B+ Trees	Yes	Yes	Full
Query Optimization	Yes	Partial	Partial
NoSQL Databases	Partial	Yes	Partial
Stored Procedures / Triggers	Partial	Yes	Partial

Table 4.6 — Comparative Coverage: B.Tech DBMS Syllabus vs. Scaler Course

The comparative analysis shows that the Scaler Topics DBMS course provides full coverage of 8 out of 11 major DBMS topic areas in the standard B.Tech curriculum. Partial coverage exists for concurrency control (the protocol depth is lighter than what some universities require), query optimization (conceptual coverage without hands-on cost-based optimization), and stored procedures / triggers (covered in later modules but with less depth than a DBA-focused curriculum would provide).

Chapter 5 — Discussion & Recommendations

5.1 Strengths of the Course

The Scaler Topics DBMS course demonstrates several significant strengths that make it a highly effective free resource for second-year B.Tech students:

5.1.1 Conceptual Depth and Clarity

The video lecture quality is the course's strongest asset. Srikanth Varma's approach of building examples incrementally — starting with a simple two-table database and progressively adding complexity — mirrors best practices in constructivist pedagogy. The normalization modules (1NF through BCNF) are particularly well-structured: each normal form is introduced with a problematic table, the anomaly is demonstrated (insertion, update, deletion anomalies), and the normalization step is applied with explicit explanation of why it resolves the problem. This approach mirrors how the topic appears in placement interviews, where candidates must both identify violations and apply corrections.

5.1.2 Comprehensive SQL Coverage

The SQL modules (4–7) provide exceptional coverage of the queries and clauses most commonly tested in technical interviews and college practicals. The walkthroughs of complex JOINs (particularly SELF JOINs and nested subqueries) use concrete table examples with visible data, which makes the logical operations far clearer than abstract descriptions. Students in the survey consistently cited the SQL modules as the most immediately useful for their academic coursework.

5.1.3 Alignment with Placement Assessment Topics

The course is explicitly designed with technical interview preparation in mind, which aligns closely with the priorities of second-year B.Tech students who begin preparing for campus placements by their third semester. This means that topics like ACID properties, transaction isolation levels, B+ tree indexing, and normalization — all high-frequency interview topics — receive disproportionately thorough treatment compared to more obscure DBMS topics.

5.1.4 Free Access and Certificate

The course is completely free, with no time pressure (lifetime access, self-paced), and provides a Certificate of Excellence upon completion. This makes it uniquely positioned among MOOC offerings — delivering industry-quality instruction at zero cost, with a credential that can be added to a LinkedIn profile or resume.

5.2 Weaknesses of the Course

5.2.1 No Integrated SQL Execution Environment

The most significant limitation identified in both personal evaluation and survey feedback is the absence of an embedded SQL sandbox within the platform. Students are directed to install MySQL Workbench externally, which introduces setup friction — particularly for students on Windows systems where MySQL installation is frequently complicated by PATH configuration issues and service startup errors. This setup barrier disproportionately affects students who are new to SQL environments and may lead some to abandon practical exercises before they begin.

5.2.2 Weak Assignment Auto-Feedback

The 16 in-module challenge sets provide correct answers for MCQs but do not explain why incorrect options are wrong. For SQL-based challenges, there is no query validator — students receive no feedback on whether their query approach is optimal, even if they produce the correct output through an inefficient method (e.g., using a correlated subquery where a JOIN would be more efficient). This limits the assignment component's effectiveness as a learning tool beyond simple factual recall.

5.2.3 Limited Real-World Case Studies

The course uses a consistent but generic 'University' database example throughout. While consistency aids comprehension, students expressed a desire for at least one module that applies DBMS concepts to a recognizable production system — such as the schema design for an e-commerce platform, a social media database, or a banking system. Such examples would make the relevance of DBMS concepts more viscerally clear and better prepare students for the kind of system design discussions that appear in advanced placement rounds.

5.2.4 Concurrency Control Depth

Modules 11–12 on concurrency control and deadlock handling cover the conceptual material adequately but move quickly through some of the more complex protocols (e.g., Strict 2PL vs. Rigorous 2PL, timestamp-based concurrency control). Students who encounter these topics in examinations may find the course coverage insufficient for detailed problem-solving questions.

5.3 Recommendations

5.3.1 Recommendations for Video Lectures

1. Add animated ER diagram construction walkthroughs: Replace static ER diagrams with step-by-step animated construction videos for complex scenarios (e.g., hospital management system, airline reservation system). Animation makes the entity identification and relationship mapping process more intuitive than static completed diagrams.

2. Add one industry-scale case study module: Introduce a dedicated module (approximately 3–4 lessons) that designs a complete database schema for a recognizable system (e.g., an Amazon-like e-commerce platform) end-to-end — from requirements to ER diagram to normalized schema to SQL implementation. This bridges the gap between academic examples and real-world application.
3. Expand concurrency control coverage: Add 2–3 additional lessons on concurrency control protocol comparisons (Strict 2PL vs. Rigorous 2PL, timestamp vs. validation-based protocols) with worked examples and problem-solving demonstrations.

5.3.2 Recommendations for Assignments

1. Add detailed explanations for incorrect MCQ options: For every MCQ challenge, add a brief explanation (2–3 sentences) for each incorrect option explaining why it is wrong. This transforms challenges from simple recall tests into active learning opportunities.
2. Introduce auto-graded SQL challenges: Partner with a browser-based SQL engine (e.g., SQLiteOnline, Mode Analytics, or a custom Scaler sandbox) to provide auto-graded SQL challenges where students write queries and receive immediate correctness and efficiency feedback.
3. Add timed mock assessment at course end: Introduce a final 30-question timed assessment covering all 16 modules, simulating the time pressure of a placement SQL round or university examination. This provides students with a realistic measure of their preparedness.

5.3.3 Recommendations for Practicals

1. Embed a browser-based SQL sandbox: The single most impactful improvement would be integrating an in-browser SQL execution environment (similar to the approach used by W3Schools SQL Tryit, Mode Analytics, or Khan Academy's CS courses). This eliminates setup friction entirely and allows students to run queries directly within the Scaler Topics interface.
2. Add detailed setup guides for all major operating systems: Provide step-by-step MySQL Workbench setup guides with troubleshooting sections for Windows 10/11, macOS Ventura+, and Ubuntu 22.04 — covering the most common installation errors.
3. Include ER diagram design practicals: Add 2–3 practical tasks where students must design a complete ER diagram from a written requirements description, then convert it to a relational schema and implement it in SQL. This would directly mirror the type of practical examination tasks set in most university DBMS courses.

Chapter 6 — Conclusion

This minor project has undertaken a structured evaluation of the Scaler Topics "DBMS Course: Master the Fundamentals and Advanced Concepts" from the perspective of a second-year B.Tech CSE student enrolled in a formal DBMS course. The evaluation combined personal completion of the full 29-hour course with a structured Likert-scale survey of 35 second-year peers, producing a comprehensive assessment across three dimensions: video lectures, assignments, and practicals.

The findings are unambiguously positive at the overall level — 88.6% of survey respondents rated the course as Good or Excellent, and the overall mean rating of 4.2 / 5 places it firmly in the "highly recommended" category for supplementary DBMS learning. The video lectures, with a mean rating of 4.3 / 5, represent the course's strongest component: the production quality, conceptual clarity, and systematic instructional approach of Srikanth Varma's teaching make complex DBMS topics accessible and engaging.

The assignment component (3.9 / 5) is effective but limited by the absence of detailed feedback on incorrect answers and the lack of auto-graded SQL query validation. The practical component (3.6 / 5) is the weakest link — not because the practical tasks themselves are poorly designed, but because the absence of an integrated SQL execution environment creates unnecessary setup friction that deters some students from completing the hands-on exercises.

From a curriculum alignment perspective, the course provides full coverage of 8 of 11 major B.Tech DBMS topics, with partial coverage of concurrency control depth, query optimization, and stored procedures. This coverage profile makes the course an excellent complement to formal coursework — it reinforces the core topics most thoroughly while acknowledging that some advanced areas require supplementation from textbooks or official course notes.

The three most impactful improvements the course creators could make are: (1) integrating a browser-based SQL execution sandbox to eliminate practical setup friction; (2) adding detailed explanations for incorrect MCQ options and introducing auto-graded SQL challenges; and (3) introducing one industry-scale case study module that applies all DBMS concepts to a realistic production database design scenario.

In conclusion, the Scaler Topics DBMS course is the most accessible, highest-quality free DBMS MOOC currently available for Indian B.Tech students. It is strongly recommended as a primary supplementary learning resource for second-year B.Tech CSE students — particularly for normalization, SQL, and ACID transaction concepts — and as an interview preparation tool for students targeting SDE roles. With the improvements recommended in Chapter 5, it has the potential to become a definitive free resource for university-level DBMS education in India.

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Appendices

Appendix A — Survey Questionnaire

SECTION A: Video Lectures

(Rate each item: 1 = Very Poor, 2 = Poor, 3 = Average, 4 = Good, 5 = Excellent)

16. How clear were the explanations of DBMS concepts (ER modelling, SQL, normalization, ACID)?
17. How appropriate was the pacing and length of individual video lessons?
18. How effective were the visual aids used (ER diagrams, SQL walkthroughs, transaction diagrams)?
19. How would you rate the overall audio and video production quality?
20. How engaging was the instructor's teaching style?
21. How thoroughly did the course cover topics relative to your B.Tech DBMS syllabus?
22. How useful were the real-world examples and applications provided?
23. Overall, how satisfied are you with the video lecture component?

SECTION B: Assignments / In-Module Challenges

24. How relevant were the challenges to SQL, normalization, and transaction theory?
25. How appropriate was the difficulty level of challenges for your academic level?
26. How clear were the challenge instructions?
27. How helpful was the feedback / answer explanation provided after each challenge?
28. How did the time required for challenges compare to the learning value derived?

SECTION C: Practicals

29. How valuable were the hands-on SQL and ER modelling practical tasks?
30. How usable was the recommended tool / environment for practical exercises?
31. How easy was it to set up the required software / tools?
32. How adequate was the troubleshooting support available within the course?
33. Overall, how satisfied are you with your practical learning outcome?

SECTION D: Overall Evaluation

34. Overall course rating (1–5):
35. Is this course suitable for second-year B.Tech DBMS learning? (Yes / Partially / No)
36. Open-ended: What is the single most significant improvement you would suggest?

Appendix B — Raw Data Summary (n = 35 Responses)

Parameter	Mean	SD	5 (Exc.)	4 (Good)	3 (Avg)	2 (Poor)	1 (V.Poor)
Video Lecture Clarity	4.5	0.52	16	16	3	0	0
Pacing & Length	4.1	0.61	9	20	5	1	0
Visual Aids	4.3	0.58	13	17	4	1	0
AV Quality	4.6	0.49	18	15	2	0	0
Engagement	4.4	0.55	14	18	2	1	0
Syllabus Depth	4.2	0.60	11	18	5	1	0
Real-World Examples	3.9	0.72	8	16	8	3	0
Overall Lectures	4.3	0.57	13	17	4	1	0

Table B.1 — Raw Data Summary: Video Lecture Parameters

Appendix C — Selected Student Comments (Anonymized)

The following are representative open-ended responses from survey respondents (anonymized by student ID):

Student 04: "The normalization modules are excellent — I had struggled with BCNF for weeks from my college notes and this course explained it better in 20 minutes than my lectures did in two weeks."

Student 11: "Please add a built-in SQL editor. I spent 45 minutes trying to install MySQL Workbench before giving up and using an online tool instead. The practical content itself was good once I could actually run queries."

Student 17: "The instructor's style is very clear. But I wish there were examples from real companies — showing the database schema of a food delivery app or an e-commerce site would make the theory feel much more relevant."

Student 22: "Best free DBMS course I have found. I would compare it favourably to paid courses on Udemy. The certificate is a bonus — I have already added it to my LinkedIn."

Appendix D — Course Completion Details

- Course: DBMS Course — Master the Fundamentals and Advanced Concepts
- Platform: Scaler Topics (scaler.com/topics)
- Instructor: Srikanth Varma
- Completed by: Roopesh Singhal (RA2411030030042)
- Completion Date: 10 April 2026
- Certificate: Certificate of Excellence — Scaler Topics
- Modules Completed: 16 / 16

- Lessons Completed: 90 / 90
- Challenges Completed: 16 / 16
- Total Time Spent: Approximately 31 hours (including re-watching selected lessons and completing SQL practice exercises)

Appendix E — List of Abbreviations (Expanded)

Abbreviation	Full Form
DBMS	Database Management System
SQL	Structured Query Language
DDL	Data Definition Language
DML	Data Manipulation Language
ER	Entity-Relationship (diagram / model)
MOOC	Massive Open Online Course
LMS	Learning Management System
ACID	Atomicity, Consistency, Isolation, Durability
1NF	First Normal Form
2NF	Second Normal Form
3NF	Third Normal Form
BCNF	Boyce-Codd Normal Form
B+ Tree	Balanced Plus Tree (index structure)
FD	Functional Dependency
2PL	Two-Phase Locking Protocol
NPTEL	National Programme on Technology Enhanced Learning
SDE	Software Development Engineer
B.Tech	Bachelor of Technology
SRM IST	SRM Institute of Science and Technology
CSE	Computer Science and Engineering